114-3海攬班高二上重補修_學習單

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Topic 1: Advanced 1D Kinematics

Q1-1

(a) If an object moves steadily from **MATH_0** to **MATH_1** in **MATH_2**, what is its average velocity? (b) Draw the Position-Time (x-t) graph for this object, and find its position at **MATH_3**. (c) From your x-t graph in (a) and (b), calculate the "slope" for the time intervals **MATH_4** and **MATH_5**. What physical quantity does the slope represent? What does a constant (same) slope mean?

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- Explanation:

Q1-2

(a) A car's velocity increases steadily from **MATH_6** to **MATH_7** in **MATH_8**. What is its average acceleration? (b) Draw the Velocity-Time (v-t) graph for this car. What is its velocity exactly at **MATH_9** ? (c) Calculate the "slope" of this v-t graph. What physical quantity does this slope represent? Also, calculate the "area under the curve" from **MATH_10** to **MATH_11**. What physical quantity does this area represent?

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Q1-3: An airplane accelerates down a runway from rest (**MATH_12**) at a constant rate of **MATH_13** for $10, \mathrm{s}$. Here are three main kinematic equations: (1) $v = v_0 + at$ (2) **MATH_15** (3) **MATH_16** (a) Choose the formula that you need, plug in the values, and calculate its final velocity just before takeoff. (b) Now, solve the exact same problem by drawing a Velocity-Time (v-t) graph for this airplane. Explain how you find the final velocity at **MATH_17** using only the graph (without plugging numbers into the formula).

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- Explanation:

Q1-4: Using the same airplane from Q1-3 (acceleration = **MATH_18**, time = **MATH_19**, initial velocity = **MATH_20**), we want to know the distance it traveled. (a) Based on three main kinematic equations in Q1-3, choose the correct formula, plug in the values, and calculate how far the airplane traveled. (b) Now, solve the exact same problem by looking at the Velocity-Time (v-t) graph you drew in Q1-3.

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- Explanation:

Q1-5: A car is traveling at **MATH_21**. The driver hits the brakes, providing a constant deceleration of **MATH_22** until it comes to a complete stop (**MATH_23**). (a) Using the three kinematic equations listed in Q1-3, choose the correct formula(s) to calculate the total stopping distance (**MATH_24**). (b) Now, solve the exact same problem by drawing a Velocity-Time (v-t) graph for this braking car. Denote how you can find the stopping time, and determine the total stopping distance purely from the graph.

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Q1-6: You drop a stone from a tall bridge from rest (**MATH_25**, **MATH_26**). It takes exactly **MATH_27** to hit the water. (a) Using the three kinematic equations, choose the correct formulas to calculate TWO things: its final velocity just before hitting the water, and its displacement. (b) Now, solve the exact same problem by drawing a Velocity-Time (v-t) graph. Denote the final velocity from the graph ,and determine the displacement. Why is the area a negative number?

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- Explanation:

Q1-7: A baseball is thrown straight up into the air with an initial velocity of +15 m/s. Gravity acts on it with **MATH_28**. At its maximum height, its instantaneous velocity is exactly **MATH_29**. (a) Using the three kinematic equations, choose the correct formula to calculate the maximum height it reaches. (b) Now, solve the exact same problem by drawing a Velocity-Time (v-t) graph. (hint: The line will start at v = +15 and have a negative slope.)

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Q1-8: A sprinter runs in a two-stage motion.

(a) Stage 1: A sprinter starts from rest (0 m/s) and accelerates at 2 m/s^2 for 4 s. What is his speed at the 4-second mark? How far did he run during these 4 seconds? (b) Stage 2: He then maintains that top speed (constant velocity) for another 6 s. How far did he run during Stage 2? (c) What is his total displacement for the entire 10 seconds? Draw a v-t graph for this 10-second run to verify your answer.

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- Explained by:
- Explanation:

Topic 2: 2D Kinematics and Projectiles

Q2-1: You are on a cliff that is **MATH_30** high. We use **MATH_31**. (a) Ball A is dropped vertically from rest. What are its initial horizontal velocity (**MATH_32**) and initial vertical velocity (**MATH_33**)? (b) Calculate the time it takes to hit the ground. (c) Ball B is thrown perfectly horizontally at a fast speed of **MATH_34**. What is its initial horizontal velocity (**MATH_35**) and its initial vertical velocity (**MATH_36**)? (c) Using the **MATH_37** you just found for Ball B, calculate the time it takes to hit the ground. (d) Compare your answers for (a) and (c). Does the horizontal speed of Ball B change its falling time?

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- Explanation:

Q2-2: Look at Ball B from Q2-1 again (cliff height = **MATH_38**, launched horizontally at **MATH_39**). (a) Once the ball leaves your hand, gravity is the only force acting on it. Which direction does gravity pull? (b) Ignoring air resistance, what is its horizontal acceleration (**MATH_40**)? What kind of motion does Ball B have in the horizontal direction? (c) Calculate the horizontal distance (**MATH_41**) Ball B traveled during its **MATH_42** in the air.

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- Explained by:
- Explanation:

Q2-3: A stunt car wants to drive off a perfectly flat, **MATH_43** high cliff and safely land on a target pad that is exactly **MATH_44** away from the base of the cliff. (a) Y-AXIS First: Using the height of the cliff, calculate how long (t) the car will be falling in the air. (b) X-AXIS Next: What must be the car's initial horizontal speed (**MATH_45**) to hit the target?

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- Explained by:
- Explanation:

Q2-4: In 2D kinematics, any diagonal movement must be broken down into horizontal (X) and vertical (Y) parts using trigonometry (SOH CAH TOA). Suppose you kick a soccer ball with an initial velocity of **MATH_46** at an angle of **MATH_47** above the ground. (a) Write down the components for the initial horizontal velocity (**MATH_48**) and vertical velocity (**MATH_49**) using **MATH_50** and **MATH_51**. (b) Calculate the exact values of **MATH_52** and **MATH_53**.

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- Explained by:
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Q2-5: Based on Q2-4, now focus ONLY on the Y-AXIS. (a) At the highest point, what's the vertical velocity (**MATH_54**)? (b) Calculate the maximum height (**MATH_55**) the ball reaches. (c) Calculate the time it takes to reach the max height. (d) Find the total time the ball is in the air.

- Reference Video: ____________________
- Explained by:
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Q2-6: Same base on Q2-4, and use the information for Q2-5. Now focus ONLY on the X-AXIS. (a) Does gravity affect the X-axis? What is the horizontal acceleration (**MATH_56**)? (b) Calculate the total horizontal distance (**MATH_57** the ball traveled.

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Q2-7: Same base on Q2-4, and use the information for Q2-5 and Q2-6. (a) It takes 2 s to reach the maximum height. How long does it take to fall from the maximum height back to the ground? (b) Right before it hits the ground, what is its horizontal and vertical velocity respectively? (c) Based on your answers in (b), what is the total speed of the ball right before it hits the ground?

- Reference Video: ____________________

- Explained by:
- Explanation: